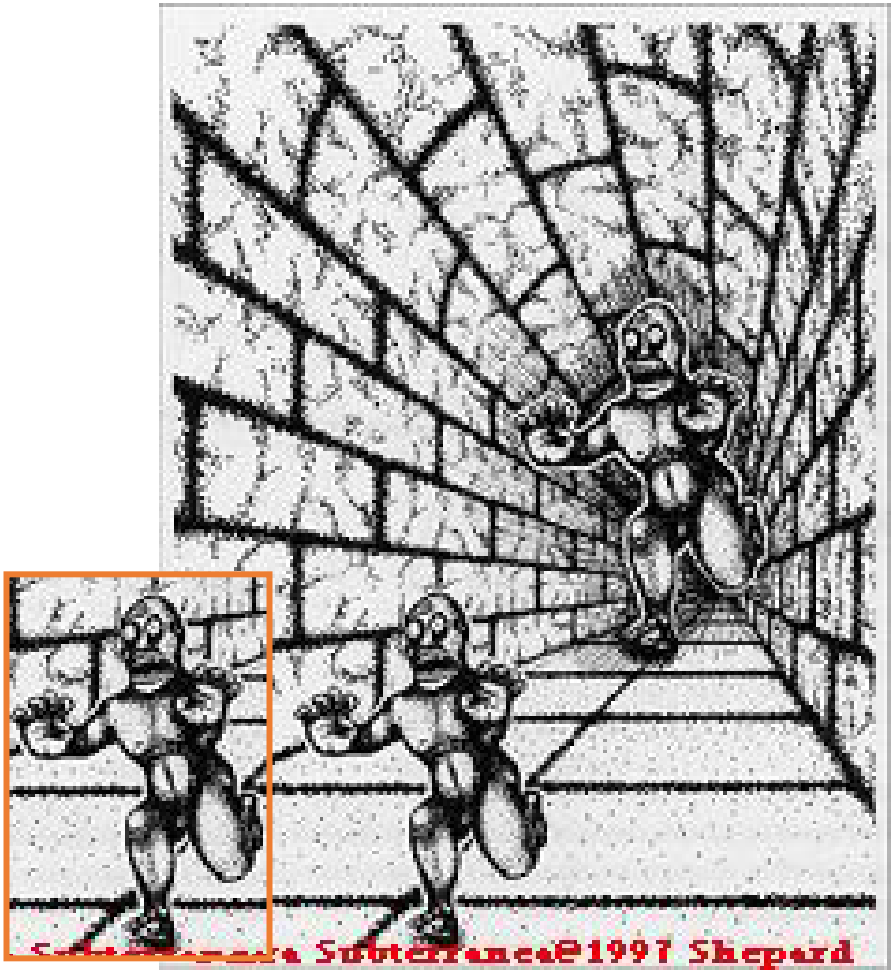
Geometry 1:
Projection and Epipolar Lines

Introduction to Computer Vision

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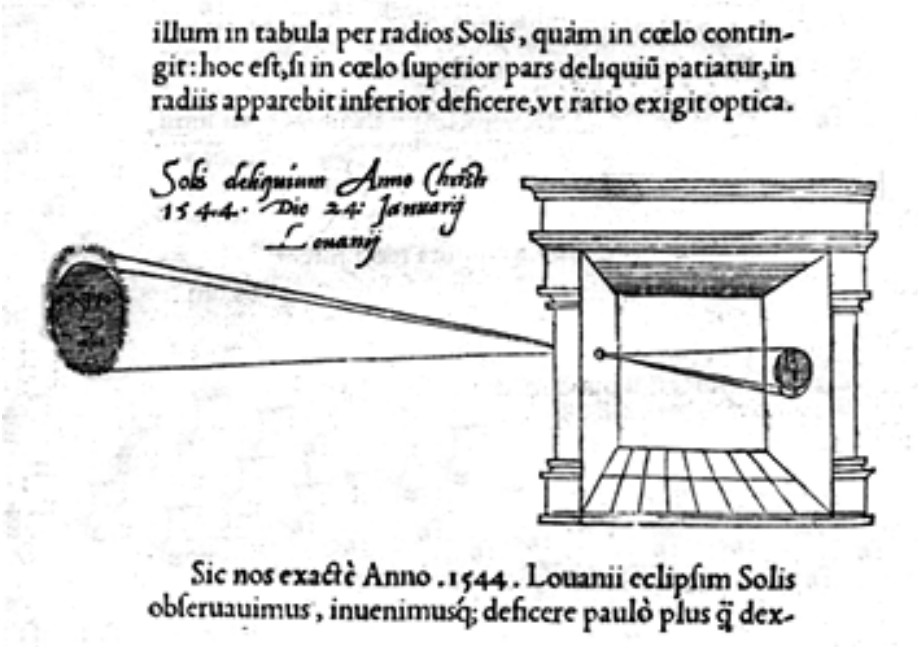
Perspectivity



Material covered

- Pinhole camera model, perspective projection
- Two view geometry, general case:
  - Epipolar geometry, the essential matrix
  - Camera calibration, the fundamental matrix
- Two view geometry, degenerate cases
  - Homography (planes, camera rotation)
  - A taste of projective geometry
- Stereo vision: 3D reconstruction from two views
- Multi-view geometry, reconstruction through factorization

Camera obscura ("dark room")

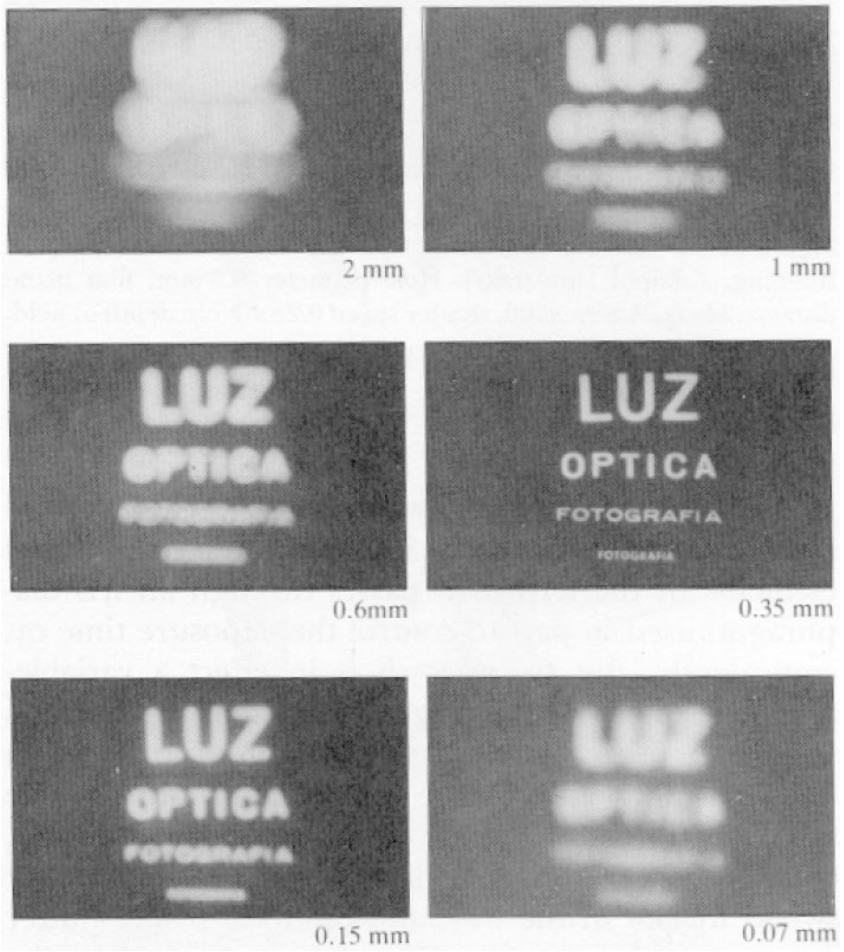


"Reinerus Gemma-Frisius, observed an eclipse of the sun at Louvain on January 24, 1544, and later he used this illustration of the event in his book De Radio Astronomica et Geometrica, 1545. It is thought to be the first published illustration of a camera obscura..."

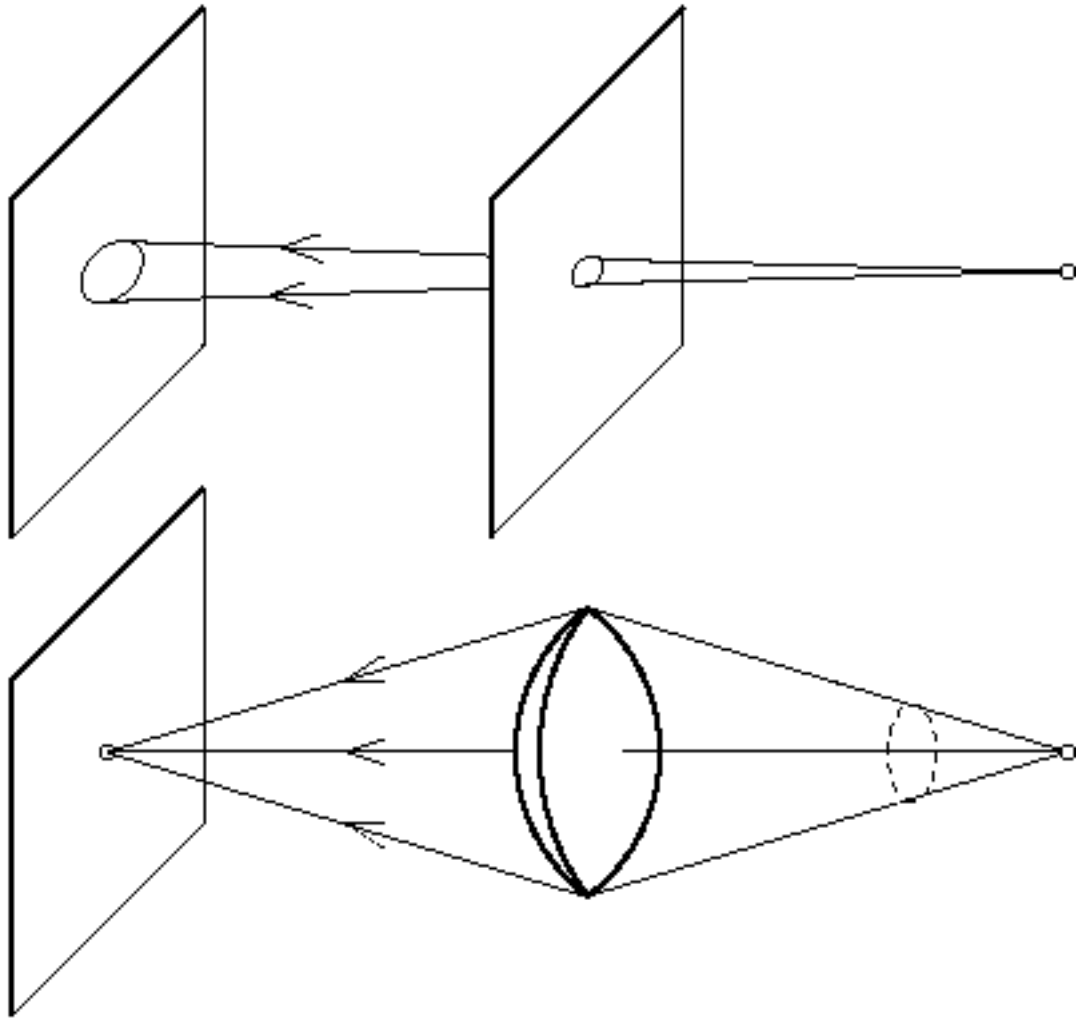
(Hammond, John H., The Camera Obscura, A Chronicle)

Why not use a pinhole camera?

- Pinhole cameras are dark
- Pinhole too big – blurry image
- Pinhole too small – diffraction



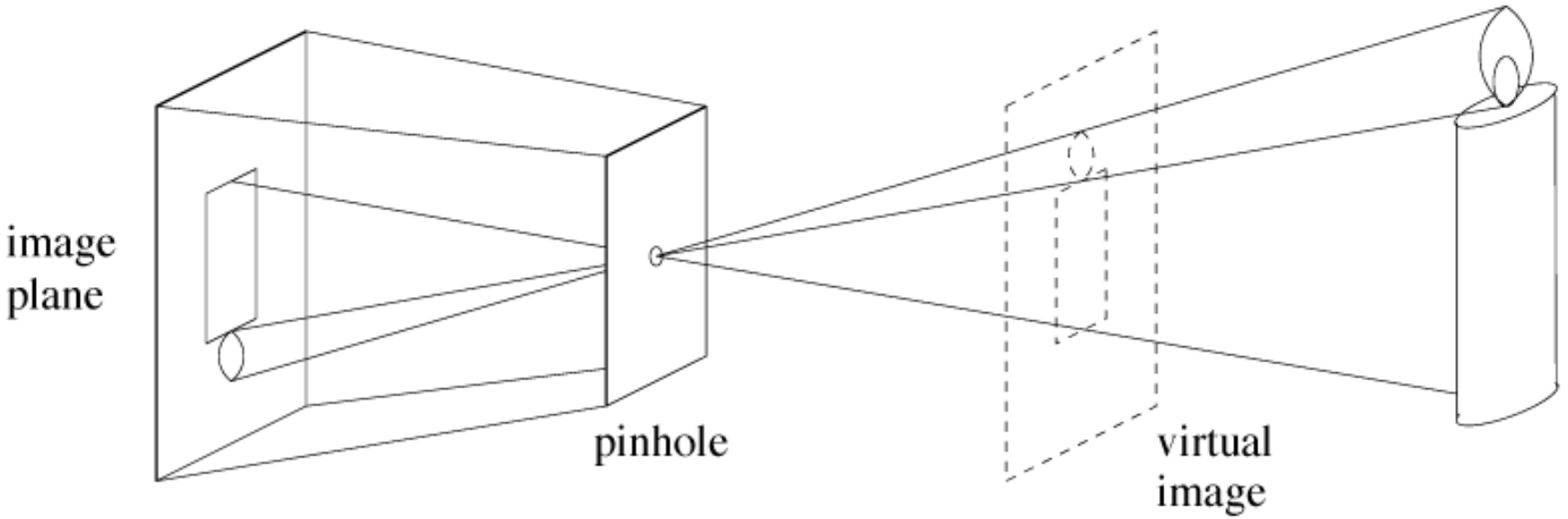
Lenses



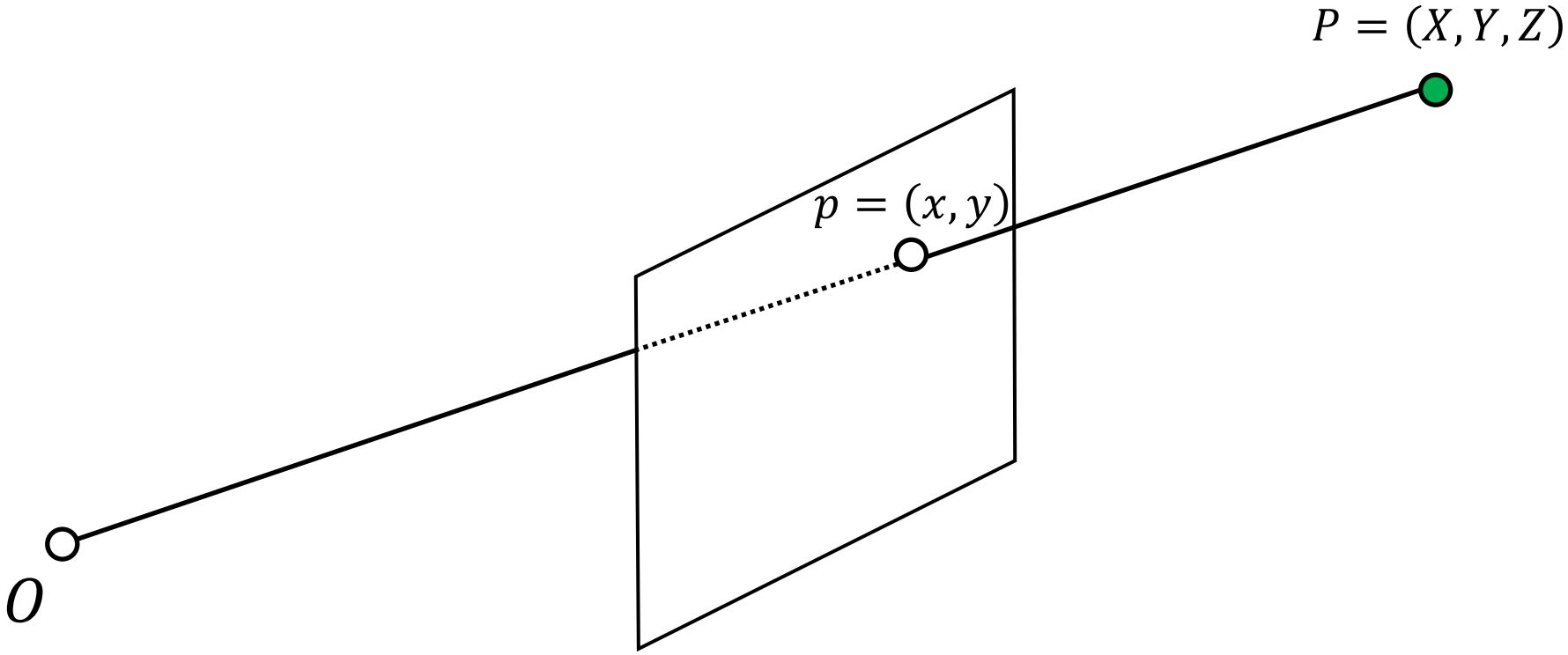
Lenses

- Lenses collect light from a large hole and direct it to a single point
- Overcome the darkness of pinhole cameras
- But there is a price
  - Focus
  - Radial distortions
  - Chromatic abberations
  - …
- Pinhole is useful as a geometric model
- Perspective: “perspicere” – to see through

Pinhole camera model

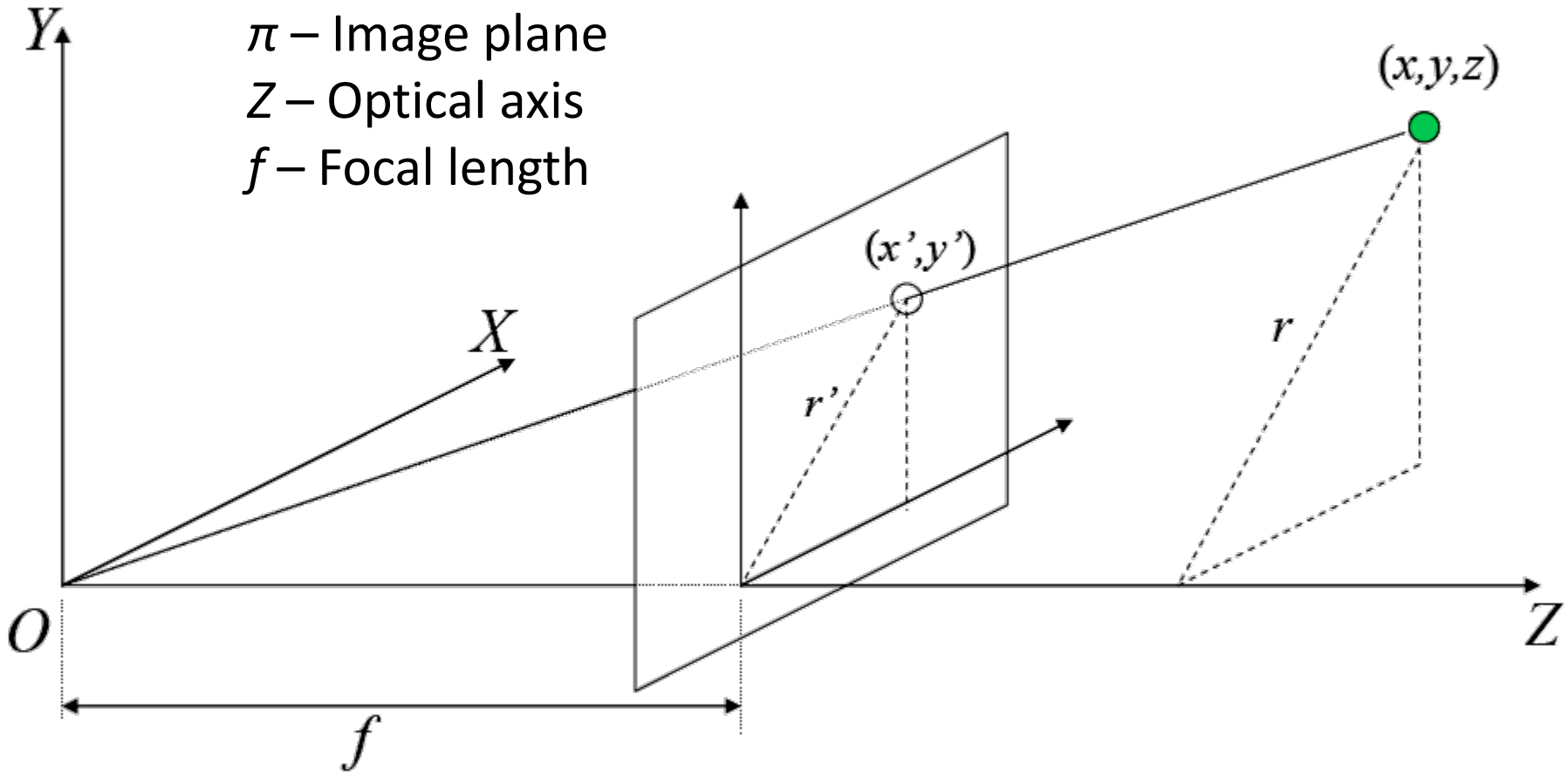


Perspective projection

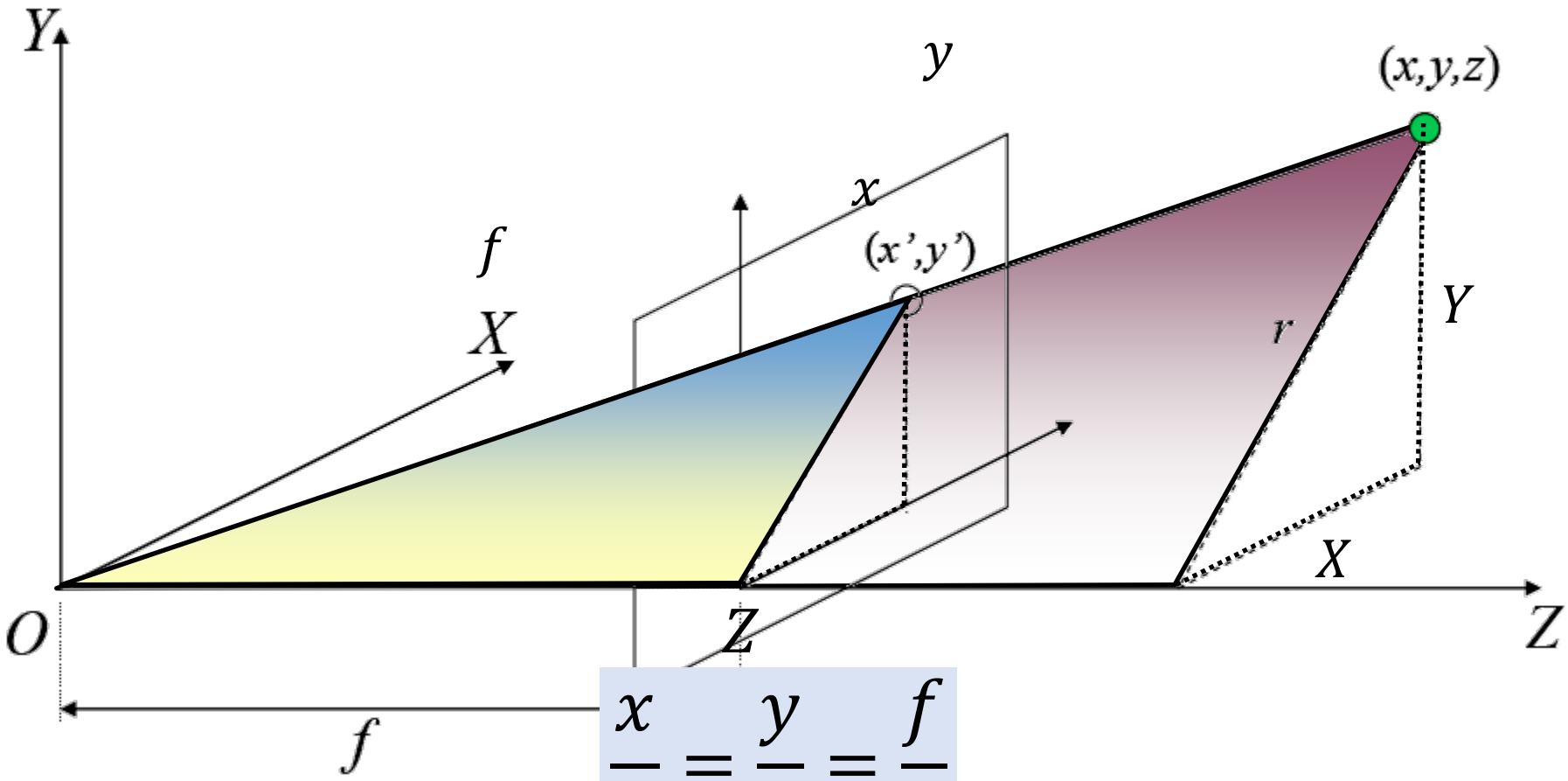


Perspective projection

O – Focal center
π – Image plane
Z – Optical axis
f – Focal length



Perspective projection



$$\frac{x}{X} = \frac{y}{Y} = \frac{f}{Z}$$

Perspective projection

- Perspective rule

$$x = \frac{fX}{Z} \qquad y = \frac{fY}{Z}$$

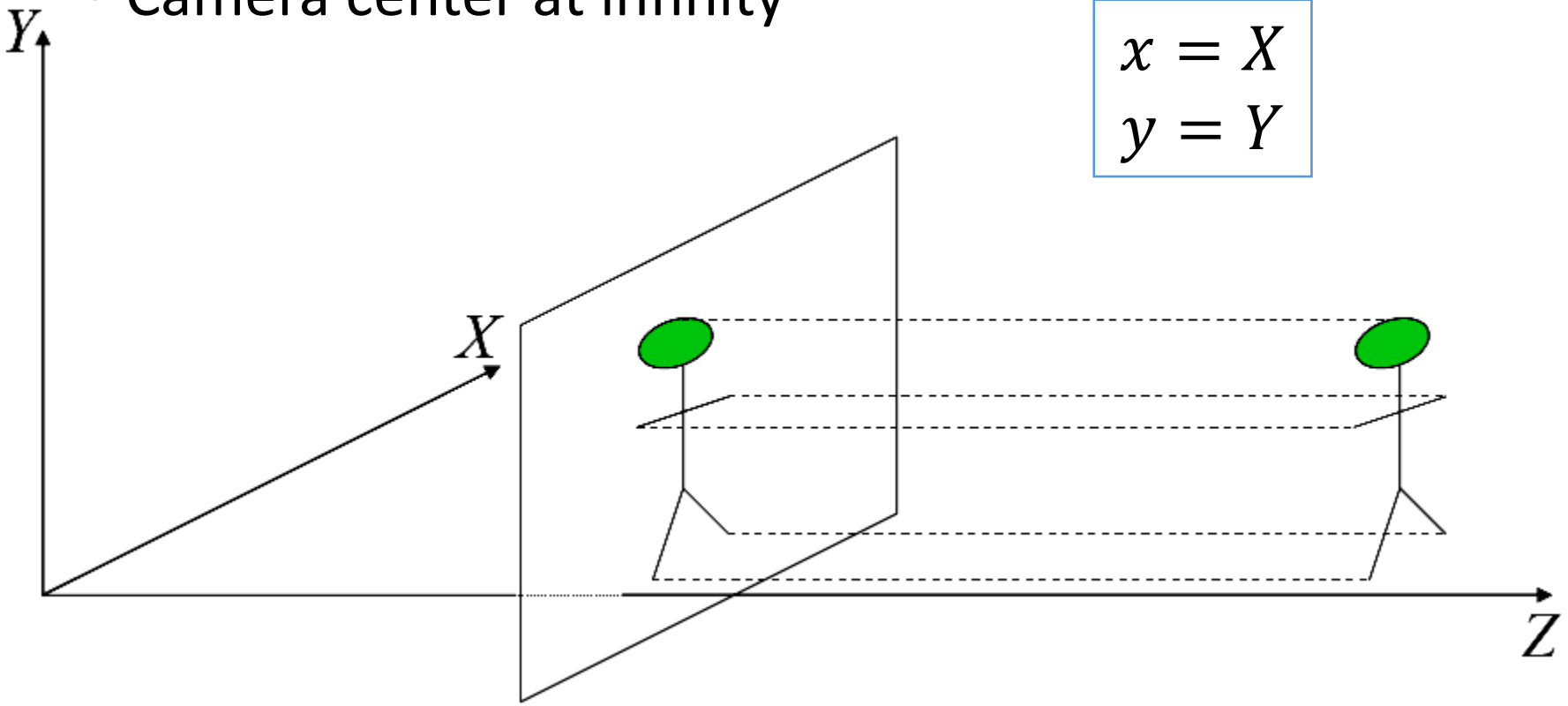
- In homogeneous coordinates

$$\begin{pmatrix} x \\ y \\ f \end{pmatrix} = \frac{f}{Z} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

Orthographic projection

- When objects are far from the camera
- Projection rays are nearly parallel
- Camera center at infinity

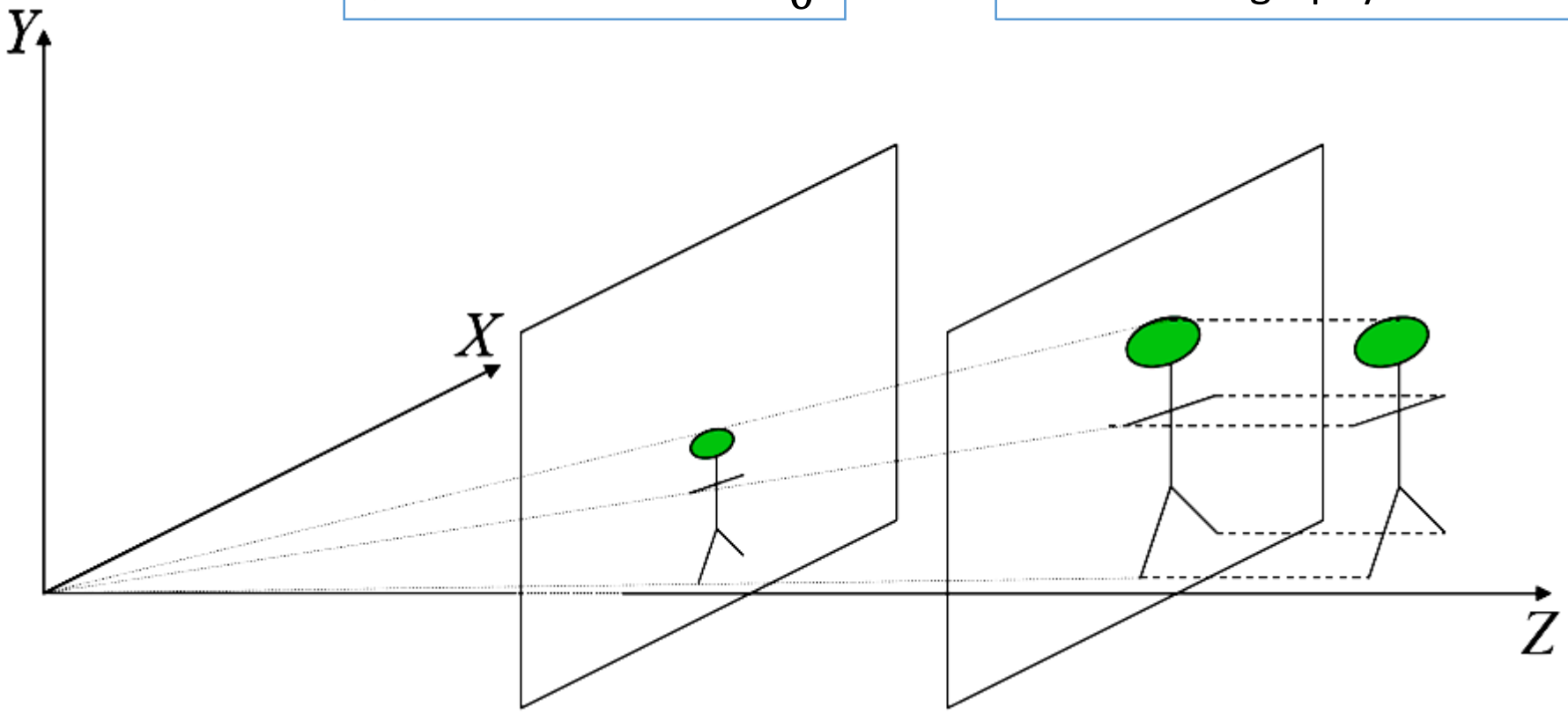
$$x = X$$
$$y = Y$$



Scaled orthographic

$$x = sX \qquad y = sY \qquad s = \frac{f}{Z_0}$$

How would a tilted rectangle look like under perspective projection? And under scaled orthography?



Which projection model should I use?

- Perspective model is needed
  - In scenes that contain many depth differences
  - For accurate 3D reconstruction (stereo, structure from motion)
- Scaled orthographic can be used
  - When objects are small relative to their distance from the camera
  - Often sufficient for recognition applications

Camera matrix

- A 3×4 matrix that captures camera location, t, orientation, R, and (linear) calibration parameters, K

$$p \propto K[R|t]P$$

$$\begin{pmatrix} x \\ y \\ f \end{pmatrix} \propto \underbrace{\begin{pmatrix} fm_x & \gamma & x_0 \\ 0 & fm_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{Internal calibration}} \underbrace{\left(\begin{array}{ccc|c} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{array}\right)}_{\text{external calibration}} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

- '$\propto$' means "up to (non-zero) scale factor." Scale is different for every point
- In "camera coordinate system" $R = I$ and $t = 0$

Calibration matrix

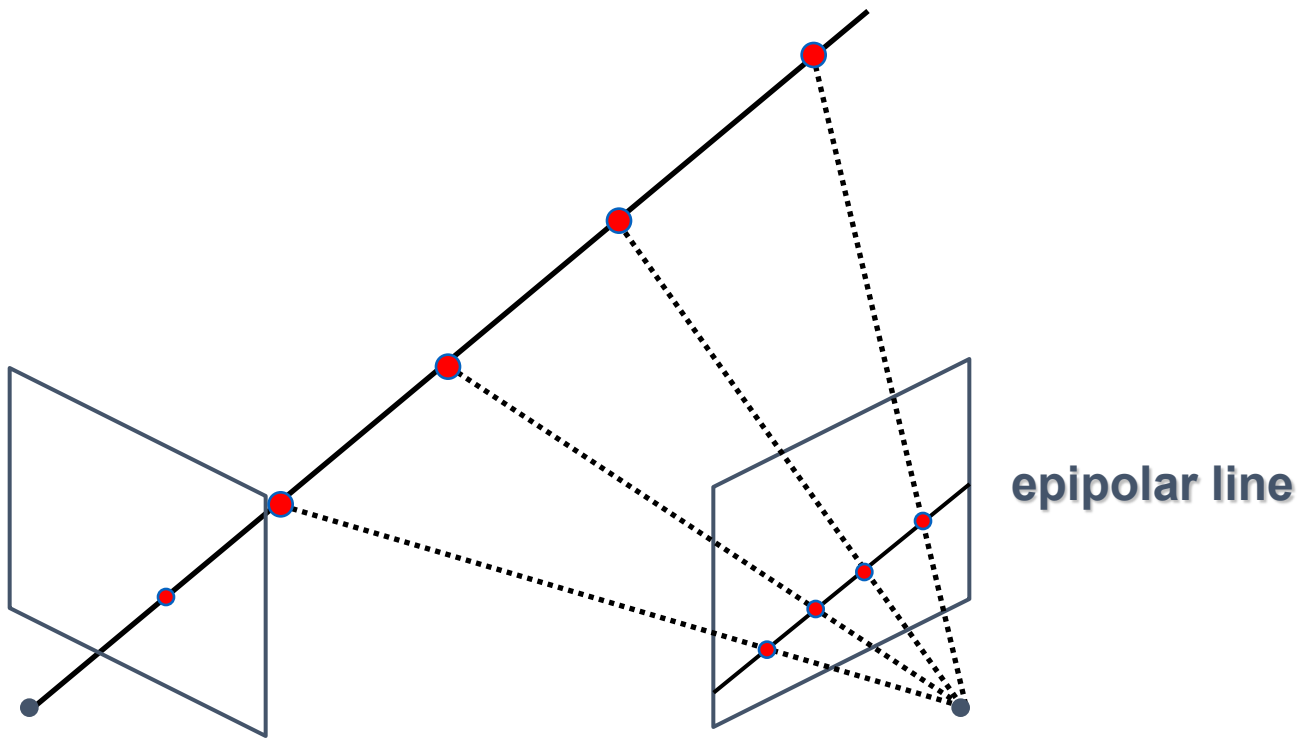
- A 3×3 upper diagonal matrix, K, that captures (linear) internal calibration parameters

$$K = \begin{pmatrix} fm_x & \gamma & x_0 \\ 0 & fm_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Parameters:
  - f - focal length
  - (m_x, m_y) - pixel size
  - γ - skew
  - (x_0, y_0) - image center
- Radial distortions are treated separately
- Both linear and radial calibration parameters are available in Exif tags

X

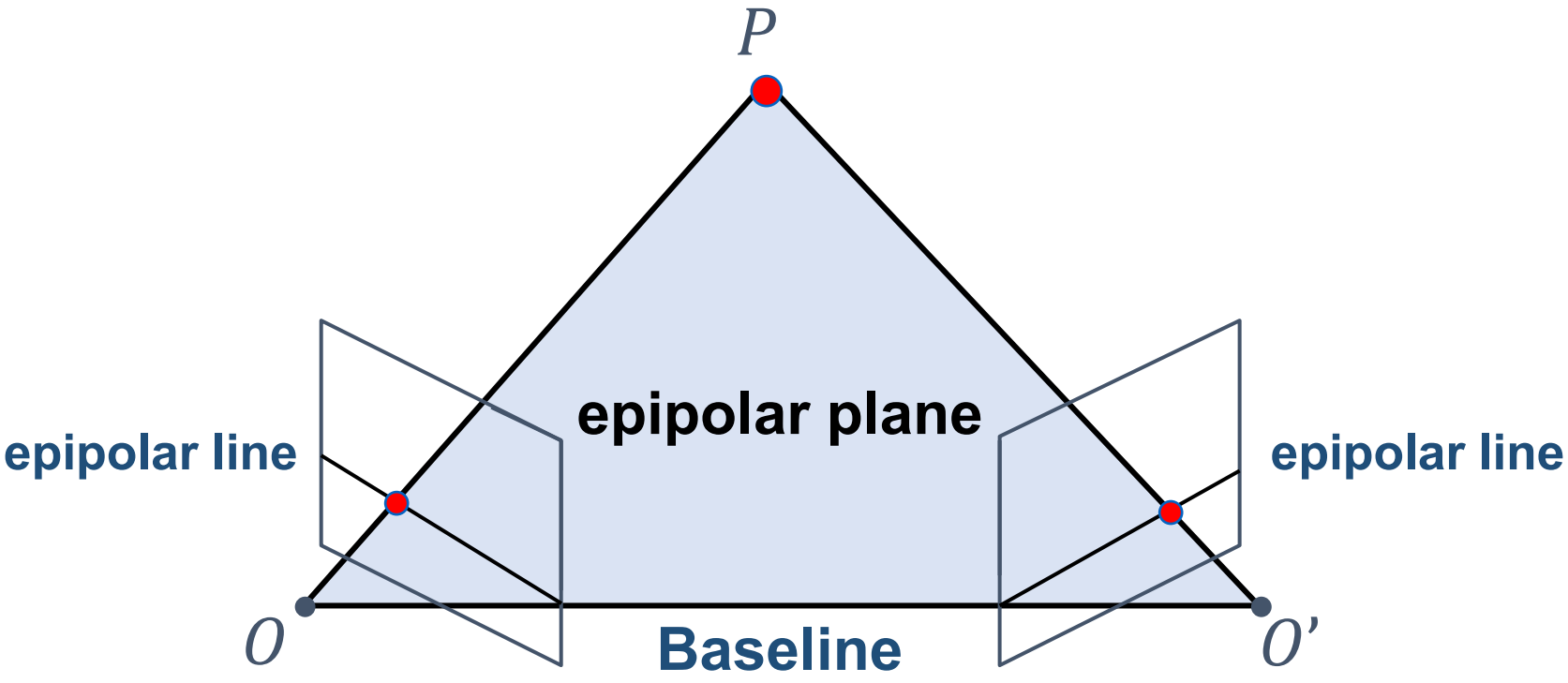
Two view geometry



Epipolar plane

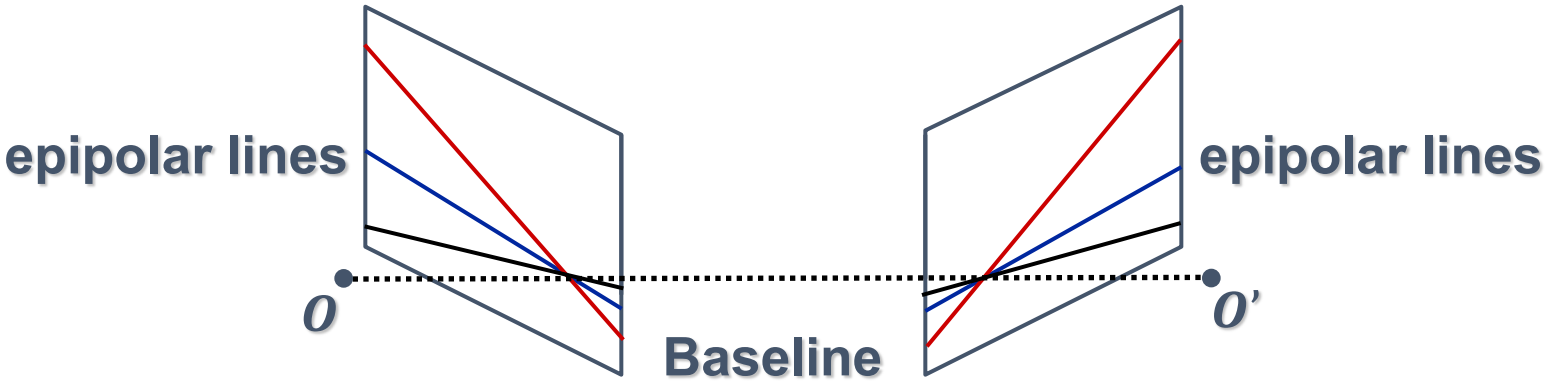
Definition:

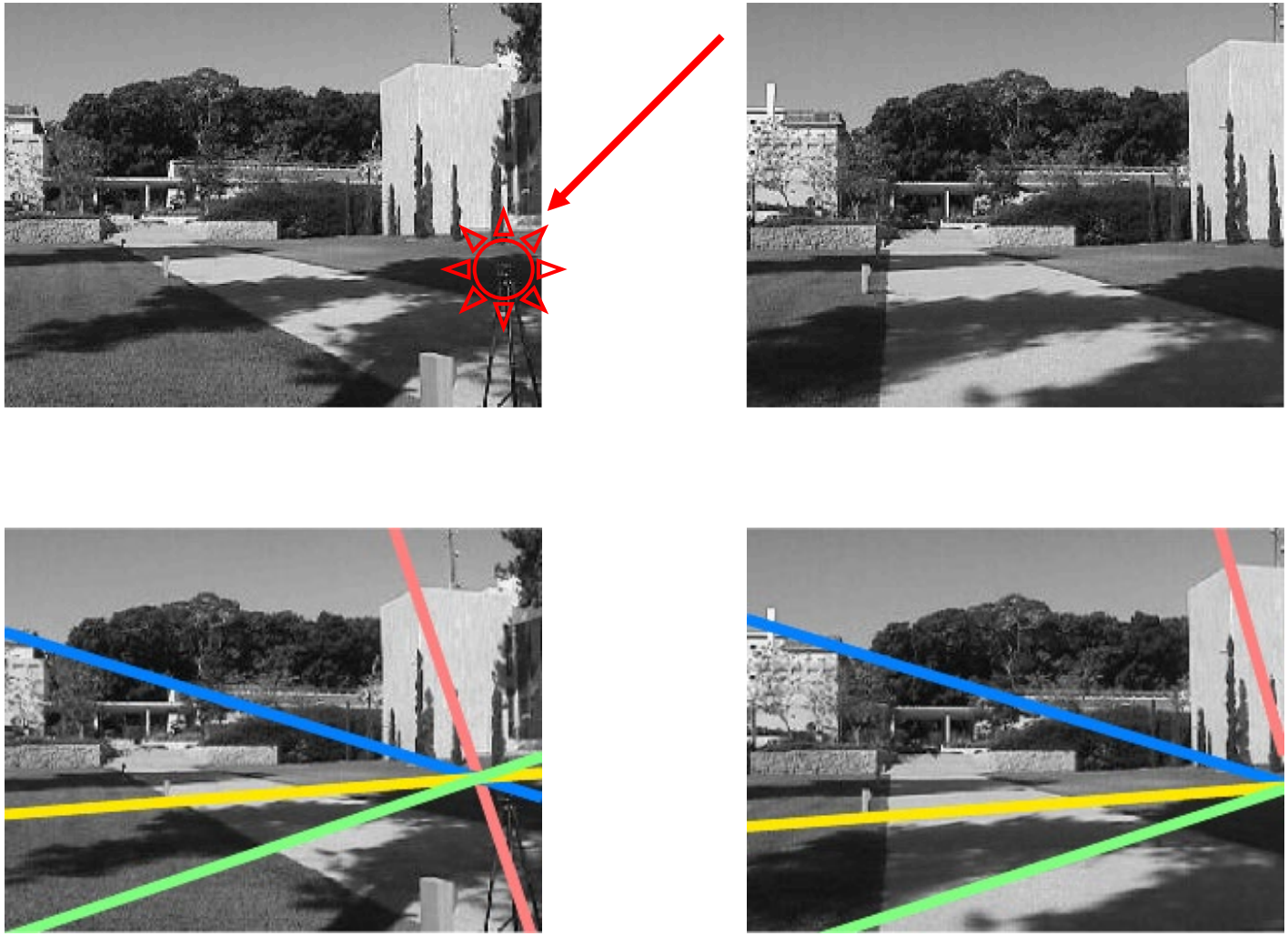
Epipolar plane: a plane that contains the baseline



Epipoles

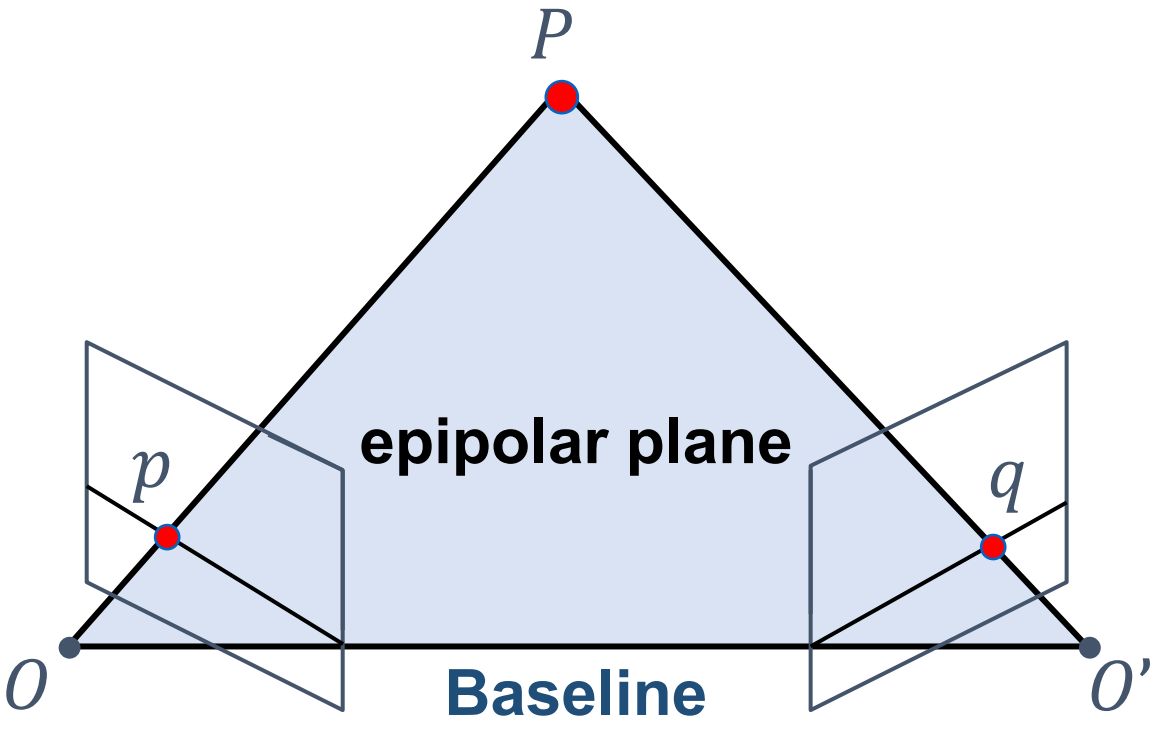
- Each epipolar plane produces a pair of epipolar lines
- There is a 1-D system of epipolar planes
- All epipolar planes contain the baseline, therefore all epipolar lines contain its intersection with the respective image planes
- These intersection points are called *epipoles*
- An epipole is the projection of the right focal center onto the left image (and vice versa)





Epipolar constraints: derivation

- We derive the constraints by requiring p, q, O, O' to lie in the same plane



Cross product, triple product

- Cross product

$$u \times v = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}$$

- $u \times v$ is orthogonal to u and v
- $|u \times v|$ is the area of the parallelogram defined by u and v
- Cross product is a linear operator expressed by a skew-symmetric matrix (verify)

$$u \times v = [u]_\times v, \text{ with } [u]_\times = \begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix}$$

- Triple product: $u^T (v \times w)$
- u, v, w are coplanar iff $u^T (v \times w) = 0$

Epipolar constraints: the Essential matrix

Assume $K = diag(f, f, 1)$ and f is known

$$p = \frac{f}{Z}P, \quad q = \frac{f}{Z}Q, \quad Q = RP + t$$

$$t \times Q = t \times RP \quad (t \times t = 0)$$

$$0 = Q^T(t \times Q) = Q^T(t \times RP) = Q^T[t]_\times RP$$

$$0 = Q^T[t]_\times RP = q^T[t]_\times Rp$$

$$\boxed{q^T E p = 0}$$

$E = [t]_\times R$ is called the *Essential* matrix

The Essential matrix

$$q^T E p = 0$$

- Given q, E defines a line (verify)
- Equation defines a necessary condition for correspondences. Is this condition sufficient?
- E is rank 2, its (right and left) null spaces contain the epipoles
- Equation is homogeneous, we can scale the scene and move cameras apart and see the same images

The Essential matrix

$$q^T E p = 0$$

- Recovery of camera position and orientation given E
  - Translation (up to scale) is given by the epipole (2 dofs)
  - Rotation can be fully determined (3 dofs)
  - 4 solutions (two rotations, sign ambiguity for translation). The correct one is found by forcing all points to have positive depths in the coordinate systems of both images
- Recovery of E (up to scale) using point matches:
  - Linear solution requires (at least) 8 matches
  - Non-linear solution requires 5 matches